

A Dual-Prediction Retrieval System

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1 Introduction

The growing prevalence of mental health concerns—such as anxiety, burnout, interpersonal stress, and low mood—has fueled rising demand for accessible, timely support resources. However, individuals often describe their experiences using informal, emotional, and highly subjective language, while most existing mental health information systems still rely on rigid clinical classification systems or keyword-based search methods. This mismatch creates a “semantic gap”: users struggle not due to a lack of resources, but because their lived experiences fail to align with how information is organized and indexed. Consequently, valuable content often remains undiscovered precisely when support is most needed.

This project addresses this issue by proposing a lightweight, non-clinical information retrieval system that maps brief, natural-language user descriptions to supportive, contextually relevant content. Rather than attempting diagnosis or treatment, the system is explicitly designed for informational support: surfacing discussion posts and guidance-style responses that users may find relatable, validating, or practically helpful. Our core hypothesis is that moderately enhancing the query understanding phase—by introducing coarse-grained mental health context prediction and simple personalization cues—can improve retrieval quality without sacrificing system transparency or robustness.

We propose a dual-prediction information retrieval framework that jointly models user input text across two dimensions: (1) the mental health-related context expressed in the text, and (2) personality or coping style tendencies inferred from linguistic cues. These signals serve not as hard constraints but as soft priors to guide subsequent retrieval and re-ranking. Specifically, the system first predicts the probability distribution of user text across broad distress categories (e.g., anxiety-

related distress, depressive signals, interpersonal stress) while concurrently predicting personality or coping style distributions derived from MBTI dimensions. These predictions are then integrated into a multi-stage retrieval process that prioritizes thematic relevance while maintaining robustness against classification uncertainty.

A key design principle is error-aware integration. In cascading NLP systems, misclassifications in early stages can severely impact downstream tasks. To mitigate this risk, we avoid hard filtering based on single prediction labels. Instead, top-ranked prediction categories and their probabilities feed into a soft re-ranking mechanism, complementing the original lexical retrieval signals. This design ensures the user’s original text remains the primary driver of retrieval, while contextual and personalized signals act as gentle biases rather than fragile gatekeeping conditions.

The system operates over a unified corpus combining discussion-style Reddit posts and structured guidance-style question-answer snippets. This design enables the system to balance peer experience narratives with more actionable guidance. Personality or coping style cues primarily adjust the relative priority of different content types in ranking, enabling limited yet interpretable personalization without relying on sensitive personal data or opaque user modeling.

We evaluate the proposed framework from both module-level and end-to-end perspectives. The classification component is evaluated using standard metrics such as Accuracy and Macro-F1, while retrieval effectiveness is measured through human-annotated relevance judgments and classic information retrieval metrics including Precision@K, nDCG@K, and Mean Reciprocal Rank. These evaluations collectively help us analyze whether introducing context-guided and lightweight personalization can tangibly enhance the relevance and utility of retrieval results.

Overall, this study demonstrates that even simple, interpretable NLP components can significantly improve user access to mental health support information when properly integrated into systems. By reframing mental health support as a retrieval and alignment problem rather than a diagnostic task, this project offers a scalable, ethical pathway for connecting individuals’ expressions of distress with relevant, appropriate support resources.

2 Related Work

2.1 Multi-Task Learning for Mental Health Detection

Social media platforms like Reddit have become important resources for studying mental health due to their anonymous nature. This encourages users to honestly disclose their psychological distress in natural language. To analyze this vast amount of unstructured data, [Cohan et al. \(2018\)](#) developed the SMHD dataset, using high-precision patterns to identify self-reported diagnoses of nine different mental health conditions. By employing a multi-task learning architecture, they demonstrated that linguistic features shared across multiple mental states could be leveraged to effectively categorize complex psychological conditions from general user input.

2.2 Mitigating Label Noise in Social Media Data

Some noise in user-generated labels, such as users posting in an incorrect subreddit, leaves a significant challenge for supervised learning models. [Haque et al. \(2021\)](#) addressed this by proposing an unsupervised label correction framework, which is designed to differentiate closely related states, such as depression and suicidal intention. Through dimensionality reduction techniques (UMAP) and Gaussian Mixture Models, they introduced a confidence-based thresholding mechanism that corrects noisy labels only when clustering confidence is high. This approach highlighted the necessity of semantic differentiation and noise robustness when dealing with ambiguous emotional contexts.

2.3 Expanding to Multi-Dimensional Wellness

More recent work has shifted the focus from clinical mental health identification toward a more general view of well-being. [Garg et al. \(2024\)](#) introduced a multi-dimensional wellness framework, ar-

guing that distress expressed in social media posts frequently spans multiple domains, such as emotional, social, and intellectual health. By implementing a transformer-based multi-label encoder equipped with a Sigmoid activation function and Binary Cross-Entropy loss, their system was able to successfully identify wellness indicators that co-exist within a single post rather than generating a single diagnosis.

3 Data Collection and Annotation

3.1 Data Sources

We use four publicly available text datasets to support both supervised classification and information retrieval. All datasets are accessed via the Hugging Face datasets library to ensure reproducibility.

Reddit mental health classification. We use [kamruzzaman-asif/reddit-mental-health-classification](#), which contains two columns: label and text. Labels correspond to subreddit-style categories (e.g., anxiety, depression, ptsd, socialanxiety, suicidewatch, as well as related domains such as relationships and divorce). We use this dataset to train the mental-health context classifier, since it provides direct supervision for mapping free-form user descriptions to coarse risk/context categories.

Reddit mental health posts for retrieval. For retrieval, we use [solomonk/reddit_mental_health_posts](#), which provides Reddit post metadata and text fields including title, body, and subreddit. This dataset supports building an indexed corpus for BM25/TF-IDF search, and the subreddit field provides a weak topical signal for category-aware reranking.

Guidance-style QA snippets. To complement discussion-style Reddit posts with more structured supportive content, we use [Srishmath/mental-health-qa-bot-dataset](#), which includes topic, input, and response. Topics include common real-life stressors (e.g., burnout, workplace stress, academic pressure, overthinking, relationship breakups). We treat each response as a guidance snippet that can be retrieved and displayed to users.

MBTI-based Personality Analysis For personality prediction, we use [kaggle-mbti-cleaned-augmented](#) which was built upon the original (MBTI) Myers-Briggs Personality Type

Dataset dataset by MITCHELL J. The dataset was augmented to improve the accuracy of NLP models by increasing the size and variety of training data. Each row contains the personality type and a collection of their posts. This dataset is proved reliable, and thus can generate good machine learning algorithms that can make accurate prediction on one’s personality traits based on what they write.

3.2 Data Acquisition

We load the following datasets programmatically using Hugging Face:

```
from datasets import load_dataset

ds_cls = load_dataset("kamruzzaman-asif/
reddit-mental-health-classification")
ds_retr = load_dataset("solomonk/
reddit_mental_health_posts")
ds_qa = load_dataset("Srishmath/
mental-health-qa-bot-dataset")
ds_per = load_dataset("Shunian/kaggle
-mbti-cleaned-augmented")
```

We will log dataset versions and maintain fixed random seeds for sampling and train/validation/test splits.

3.3 Preprocessing and Cleaning

Because Reddit-style text often includes formatting artifacts, we apply consistent preprocessing: (i) remove URLs and subreddit/user mentions (e.g., “u/...”), (ii) normalize whitespace, (iii) lowercase text for lexical baselines, and (iv) filter extremely short entries (low-content posts). For very long posts, we optionally truncate to a fixed maximum length to reduce outlier effects in both classification and retrieval.

For retrieval indexing, we construct a unified document text field:

- Reddit retrieval documents: title + “\n” + body.
- QA documents: topic + “\n” + input + “\n” + response.

3.4 Labeling and Annotation Strategy

Coarse category mapping for classification.

The classification dataset contains a fine-grained label set (subreddit-level categories). Since our system aims to predict broad mental-health-related contexts, we will map fine labels into a smaller set of coarse categories (e.g., anxiety-like distress, depressive/low mood signals, trauma-related distress, relationship stress, substance-related struggles, and

other). This mapping reduces sparsity and aligns the classifier outputs with how the retrieval component filters/boosts documents. We will report the final mapping and post-mapping class distribution in the appendix.

Weak labels for retrieval. For retrieval documents drawn from Reddit, we use the subreddit field as a weak topical tag. For QA documents, we use the provided topic as a tag. These tags enable a simple category-aware reranking rule (e.g., boosting documents whose tag matches the predicted mental-health context).

Manual relevance annotation for IR evaluation.

Retrieval quality requires relevance judgments beyond weak topical tags. We will create a small manually annotated evaluation set by sampling a set of user-like queries, retrieving the top candidates using a baseline (BM25), and grading relevance on a 0–2 scale: 0 (not relevant), 1 (partially relevant), 2 (highly relevant). These judgments will support standard IR metrics such as Precision@K, nDCG@K, and MRR.

3.5 Representative Samples

To illustrate the data diversity while avoiding identifiable content, we will include short paraphrased examples in the report. For instance:

- Anxiety-like: “I can’t stop worrying about small mistakes and it’s affecting my sleep.”
- Burnout/work stress: “I feel exhausted and numb, and I can’t get myself to start tasks.”
- Relationship stress: “I keep overthinking messages and I’m scared they’ll lose interest.”

3.6 Ethical Considerations

All data used is publicly available. We do not attempt to identify users and will avoid displaying user identifiers (e.g., author). The system is designed for informational support retrieval, not clinical diagnosis. For high-risk content (e.g., crisis-related topics), we will consider safer retrieval defaults (e.g., prioritizing guidance snippets) and include a clear disclaimer in the interface.

4 Method Description

4.1 System Overview

We propose a dual-prediction retrieval system that takes a short natural-language user description as

input and returns supportive, non-clinical information that is both (i) topically relevant to the user’s expressed mental-health context and (ii) better aligned with the user’s inferred coping style. The system is composed of two main modules: a *dual prediction* module that interprets the query, and a *personalized retrieval and reranking* module that retrieves and orders candidate documents.

4.2 Dual Prediction: Understanding the User Query

Given a user query q , we run two predictors in parallel.

Mental-health context classifier. The first predictor is a multi-class classifier trained on kamruzzaman-asif/reddit-mental-health-classification. Each instance contains a short text and a subreddit-style label (e.g., anxiety, depression, ptsd, socialanxiety, suicidewatch, relationships). Since our downstream retrieval aims to support broad, user-facing contexts, we map fine-grained labels to a smaller set of coarse categories (e.g., anxiety-like distress, low-mood/depressive signals, trauma-related distress, relationship/interpersonal stress, high-risk/crisis, and other). We then train lightweight baseline models (TF-IDF + Logistic Regression and/or Linear SVM) to output a probability distribution $P(c | q)$ over the coarse categories. In the retrieval stage, we use the *top-5* predicted categories and their probabilities rather than only the top-1 label.

Personality/coping-style predictor. The second predictor estimates a coarse personality/coping tendency from the same text q . We train this model on Shunian/kaggle-mbti-cleaned-augmented, which contains MBTI labels paired with users’ text. Our baseline setting treats this as a 16-way MBTI classification task and uses TF-IDF + Logistic Regression (or Linear SVM) to output $P(s | q)$, where s is a personality/style label. Because MBTI can be fine-grained and noisy for personalization, we also consider an alternative coarse *coping-style* grouping derived from MBTI dimensions (e.g., *action-oriented/structured*, *reflective/processing*, *social-support seeking*). This coarse grouping is designed to more directly influence what kinds of retrieved content are prioritized.

4.3 Personalized Retrieval and Reranking

After obtaining the mental-health category distribution $P(c | q)$ and coping-style distribution $P(s | q)$, we retrieve and rerank supportive content from a unified corpus.

4.3.1 Corpus Construction

We construct a unified document collection $\mathcal{D}$ that combines (1) discussion-style Reddit posts from solomonk/reddit_mental_health_posts and (2) guidance-style QA snippets from Srishmath/mental-health-qa-bot-dataset.

For each Reddit document, we define

$$d_{\text{text}} = \text{title} \parallel \text{body},$$

and for each QA document we define

$$d_{\text{text}} = \text{topic} \parallel \text{input} \parallel \text{response}.$$

We store a weak topical tag for each document: subreddit for Reddit posts and topic for QA snippets.

4.3.2 Stage 1: First-stage Retrieval

We first retrieve a candidate set $\mathcal{C}$ using the original user text q as the query. Our primary lexical baselines are BM25 and TF-IDF cosine similarity using an inverted index. As an advanced option (time permitting), we will explore dense retrieval using Sentence-BERT embeddings to improve semantic matching when lexical overlap is low. This stage aims for high recall by returning the top- N candidates (e.g., $N = 100$) for reranking.

4.3.3 Stage 2: Category-guided Soft Reranking

To guide retrieval toward the user’s predicted mental-health context while reducing cascaded errors, we apply *soft* category weighting rather than hard filtering. Let $\text{TopK}(q)$ be the top- K predicted coarse categories from $P(c | q)$ (we use $K = 5$). We define a category boost term $S_{\text{cat}}(q, d)$ that increases the score of documents whose weak tag matches any predicted category, weighted by the corresponding probability. Intuitively, even if the top-1 category is slightly wrong, a correct category can still contribute via its probability mass and keep relevant documents ranked highly.

4.3.4 Stage 3: Coping-style Personalization

We further adjust the ranking using the predicted coping style. Our personalization mechanism is

intentionally simple and interpretable: we prioritize documents whose format and tone match the inferred preference (e.g., action-oriented users may benefit more from structured tips and step-by-step guidance, while reflective users may benefit from validation and reframing content). In practice, we implement this as a lightweight boost $S_{\text{style}}(q, d)$ applied primarily to QA snippets (which are more structured and consistently actionable), while keeping the influence on Reddit discussion posts smaller to avoid overfitting to noisy style heuristics. This design choice keeps the personalization feasible while still testing whether coping-style cues can improve perceived usefulness.

4.3.5 Final Scoring Function

We combine retrieval, category, and coping-style signals in a single ranking score:

$$S(q, d) = \alpha S_{\text{retr}}(q, d) + \beta S_{\text{cat}}(q, d) + \gamma S_{\text{style}}(q, d),$$

where S_{retr} is the first-stage retrieval score (BM25/TF-IDF, optionally dense similarity), and α, β, γ are tunable weights. We set α as the dominant term to ensure the original query remains the primary driver, with category and style acting as reranking priors.

5 Evaluation Method

5.1 What we evaluate

Our system contains two predictors (mental-health context + personality/coping tendency) and one retrieval-and-ranking module. Accordingly, our evaluation focuses on three questions: (1) Can the mental-health classifier assign a reasonable coarse category to a user’s text? (2) Is the personality/coping prediction stable enough to be used for personalization? (3) Does adding category guidance and simple personalization improve retrieval relevance and usefulness?

To answer these, we report both **module-level results** (classification performance) and **end-to-end results** (final ranked retrieval outputs).

5.2 Data splits and experimental setup

For both classification tasks, we use a fixed random seed to split data into train/validation/test (e.g., 80/10/10, or the dataset’s default split if provided). We train on the training set, use the validation set for model/parameter selection, and report final numbers on the test set.

For retrieval, we build a unified corpus of Reddit discussion posts and QA-style guidance snippets, and construct an inverted index to support BM25 and TF-IDF cosine similarity as our main baselines. This ensures the full pipeline is reproducible and can be compared across different reranking settings.

5.3 Evaluation 1: Mental-health context classifier

The mental-health classifier predicts a coarse distress/context category from a user query. We report:

- **Accuracy** (overall correctness),
- **Macro-F1** (robust to class imbalance),
- **Per-class Precision/Recall/F1**

Since our downstream reranking uses the top-5 category probabilities, we also report an optional but informative **Top-5 accuracy** (whether the correct category appears in the top five predictions).

5.4 Evaluation 2: Personality / coping-style predictor

The personality component is mainly used to adjust ranking preferences, so we evaluate two settings:

- **16-way MBTI classification** (exploratory baseline),
- **coarse coping-style labels** (e.g., action-oriented vs reflective vs social-support seeking), as the main personalization signal.

We report Accuracy and Macro-F1 for both. If the 16-way MBTI task is unstable on short inputs, we prioritize the coarse coping-style setting for end-to-end personalization.

5.5 Evaluation 3: End-to-end retrieval quality

Weak tags such as subreddit (Reddit) and topic (QA) are not ground-truth relevance labels, so we create a small human-judged evaluation set for IR metrics.

Query set (user-like queries). We construct a set of 50–100 user-like queries that cover multiple coarse distress categories (e.g., anxiety, burnout, relationship stress). Queries come from (1) paraphrased samples from the classification dataset (de-identified), and (2) short scenario-based queries written by the team.

Human relevance judgments. For each query, we retrieve the top- M candidates (e.g., $M = 20$) using BM25, then manually grade each (query, document) pair on a 0–2 scale: 0 = not relevant / not helpful, 1 = partially relevant, 2 = highly relevant and helpful. These judgments enable standard IR metrics.

IR metrics. Using the graded set, we report:

- **Precision@K** ($K = 3/5/10$),
- **nDCG@K** (uses 0/1/2 graded relevance),
- **MRR** (how quickly the first relevant result appears).

Compared methods combination We evaluate four end-to-end retrieval variants to isolate the contribution of each component:

- **BM25**: first-stage BM25 ranking only;
- **BM25 + Cat**: BM25 followed by category-guided soft reranking using $P(c | q)$;
- **BM25 + Style**: BM25 followed by coping-style reranking using $P(s | q)$;
- **BM25 + Cat + Style**: full system combining both reranking signals.

References

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